THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET B

Chapter 2: **Power Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $3^4 \times 3^6$ equals: 1

(a) 3^{-2} **(b)** 3^{10} **(c)** 9^{10} **(d)** 3^{24}

2. $2^9 \div 2^4$ equals: 1

(a) 2^{13} **(b)** 2^2 **(c)** 2^5 **(d)** 2^{36}

3. $11^7 \div 11^7$ equals: 1

(a) 11^{14} **(b)** 1 **(c)** 11^1 **(d)** 11^{49}

4. $(10^3)^5$ equals: 1

(a) 10^{243} **(b)** 30^5 **(c)** 10^8 **(d)** 10^{15}

5. The value of 10^{-3} is: 1

(a) -1000 **(b)** 1000 **(c)** $\frac{1}{1000}$ **(d)** $\frac{1}{30}$

6. $(2 \times 3)^4$ equals: 1

(a) $2^4 \times 3^4$ **(b)** $2^4 + 3^4$ **(c)** 6^8 **(d)** 2×3^4

7. The number 0.0035 in standard (scientific) form is: 1

(a) 3.5×10^3 **(b)** 3.5×10^{-3} **(c)** 3.5×10^{-4} **(d)** 35×10^{-3}

8. The number 0.0412 in standard (scientific) form is: 1

(a) 4.12×10^{-3} **(b)** 4.12×10^2 **(c)** 41.2×10^{-2} **(d)** 4.12×10^{-2}

9. 2^{10} is closest to: 1

(a) 10000 **(b)** 2000 **(c)** 100 **(d)** 1000

10. $\frac{5^6}{5^8}$ equals: 1

(a) $\frac{1}{25}$ **(b)** 25 **(c)** -25 **(d)** 5^{-14}

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Simplify $(3^2)^4 \div 3^6$. 2

12. Evaluate $(-1)^{101}$ and $(-1)^{100}$. 2

13. Find the value of $4^0 + 4^{-1}$. 2

14. State the law for multiplying powers with the same base. Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Simplify: $\dfrac{3^7 \times 3^3}{3^4}$. **3**

16. Simplify: $\dfrac{5^6 \times 5^2}{5^3}$. **3**

17. Simplify: $\dfrac{4^5 \times 4^3}{4^2}$. **3**

OR

Simplify: $\dfrac{6^4 \times 6^4}{6^5}$. **3**

18. Simplify: $\dfrac{2^8 \times 2^3}{2^4}$. **3**

OR

Simplify: $\dfrac{3^8 \times 3^6}{3^2}$. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Explain why $2^{-3} = \frac{1}{8}$. **5**

(ii) Simplify $\dfrac{2^{-3} \times 3^2}{6^{-1}}$.

OR

(i) The Milky Way is about 10^5 light-years wide and one light-year is about 9.46×10^{12} km. **5**
Estimate its width in km.

(ii) Express in standard form.

20. (i) Evaluate $(5^0 + 4^{-1}) \times 2^2$. **5**

(ii) Find x if $\left(\frac{3}{5}\right)^x \times \left(\frac{3}{5}\right)^{-4} = \left(\frac{3}{5}\right)^5$.

OR

Case study: The population of India is about 140 crore and area is about 32.9 lakh km². **5**

(i) Write both in standard form.

(ii) Find the approximate population density.

(iii) Compare with a city having density 11000 per km².

END OF QUESTION PAPER — Total: 40 Marks

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